ALOK SINGH

Computer Science Engineering (AI & ML)

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PROFILE

Computer Science student specializing in AI/ML engineering, with hands-on experience across the ML lifecycle and enterprise backend systems. Background spans predictive modeling, explainable AI, and scalable pipeline design, alongside applied internship experience in API-driven services and knowledge graph-based data systems using Python.

EDUCATION

B.E. in Computer Science Engineering (AI & ML) *2023 – 2027*
Viva Institute of Technology

SKILLS

Programming & Backend: Python, SQL, FastAPI, Asyncio, REST APIs, PostgreSQL
Machine Learning: Scikit-learn, Regression, Classification, Clustering
Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Power BI (Basic)
Knowledge Systems: Knowledge Graphs, NetworkX, Ontology Resolution, Semantic Layering
Tools: Git, VS Code, Jupyter Notebook, Postman

EXPERIENCE

Software Engineer Intern — Benchmark Computer Solutions Limited *05/26 – Present*
Mumbai, India

- Contributed to the development and stabilization of an enterprise-grade AI-driven data workspace platform, working across multiple modular Python services.
- Debugged and optimized backend service startup and runtime reliability, resolving environment and dependency issues across a multi-service codebase.
- Built and tested FastAPI endpoints supporting enterprise query pipelines, integrating knowledge graph traversal and ontology resolution logic for structured data retrieval.
- Worked on semantic layering for a text-to-SQL system, improving query interpretation by combining schema descriptions with example-driven retrieval techniques.

Technologies: Python, FastAPI, SQL, NetworkX, Git, REST APIs, Knowledge Graphs, Semantic Layering

PROJECTS

LegalOS — AI Legal & Compliance Operating System

- Architected and built a production-grade multi-tenant AI system using FastAPI, React 18, PostgreSQL, Qdrant, and Google Gemini API to automate contract risk scoring, legal notice drafting, and compliance tracking for Indian SMEs.
- Engineered an end-to-end RAG pipeline using local SentenceTransformers (all-MiniLM-L6-v2) and Qdrant vector database with document-level payload filtering, eliminating cross-document context leakage during legal clause negotiation suggestions.
- Implemented an async non-blocking background task pipeline in FastAPI to offload document parsing (PyMuPDF/docx), vector embedding, and LLM analysis, reducing API response latency by 95% during document ingestion.
- Integrated Sandbox.co.in live GSTIN verification APIs with OAuth/JWT token lifecycle management, in-memory caching, and automatic retry handling for real-time vendor fraud detection.

Technologies: Python, FastAPI, React 18, TypeScript, Qdrant, PostgreSQL, Google Gemini API, Docker, Tailwind CSS

Customer Churn Prediction System

- Built an end-to-end telecom churn prediction system using Scikit-learn Pipelines and ColumnTransformers.
- Performed EDA, feature engineering, imbalance handling, and model optimization to improve churn prediction performance.
- Integrated SHAP explainability for transparent feature importance analysis and business interpretability.
- Developed an interactive Streamlit dashboard with churn analytics, risk visualization, and automated retention recommendations.
- Improved churn recall using class weighting and probability threshold tuning techniques.

Technologies: Python, Pandas, NumPy, Scikit-learn, SHAP, Streamlit, Matplotlib, Seaborn